Linear Algebra Basics

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The primary objective of linear algebra is to study vectors and matrices and their properties.

1 Vectors and Matrices

1.1 Vectors

A vector is essentially a collection of scalars. In this course, we consider a vector as a column vector. If a vector is in $\mathbb{R}^d$, it has d entries. For example, a vector $\mathbf{v} \in \mathbb{R}^d$ is represented as:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_d \end{bmatrix}$$

The transpose of a vector $\mathbf{v}$, denoted $\mathbf{v}^\top$, transforms a column vector into a row vector. Specifically, if $\mathbf{v}$ is $d \times 1$, then $\mathbf{v}^\top$ is $1 \times d$.

1.2 Linear Independence

A fundamental concept when discussing vectors is linear independence. A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is linearly independent if no vector in the set can be written as a linear combination of the others. For example, consider two vectors $\mathbf{a}$ and $\mathbf{b}$. Another vector $\mathbf{c}$ is not linearly independent of $\mathbf{a}$ and $\mathbf{b}$ if it can be expressed as:

$$\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b}$$

where $\alpha, \beta \in \mathbb{R}$. In this case, $\mathbf{a}$ and $\mathbf{b}$ span $\mathbb{R}^2$. This means that any vector in $\mathbb{R}^2$ can be represented as a linear combination of $\mathbf{a}$ and $\mathbf{b}$.

1.3 Spanning Sets and Linear Independence

We generalize the concept of spanning. A set of vectors $\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$ spans $\mathbb{R}^d$ if any vector $\mathbf{c} \in \mathbb{R}^d$ can be written as a linear combination of these vectors:

$$\mathbf{c} = \sum_{i=1}^{n} \alpha_i \mathbf{a}_i$$

where $\alpha_i \in \mathbb{R}$ for all i. For the set to span $\mathbb{R}^d$, we require $n \geq d$. However, this spanning set may not be efficient. To achieve an efficient span, the concept of linear independence is introduced.

1.4 Linear Independence: Efficient Spanning

A set of vectors $\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$ is linearly independent if no vector in the set can be expressed as a linear combination of the others. In other words, if a linear combination of these vectors equals zero, then all the coefficients must be zero:

$$\sum_{i=1}^{n} \alpha_i \mathbf{a}_i = 0 \quad \text{implies} \quad \alpha_i = 0 \text{ for all } i$$

If this condition holds, the set is linearly independent. For instance, if $\mathbf{a}_i$ is a non-zero vector with a non-zero coefficient α_i, then separating $\mathbf{a}_i$ from the equation and dividing by α_i would contradict the assumption of linear independence. Hence, $\alpha_i = 0$ for all i.

2 Basis and Subspaces

2.1 Basis

A basis for $\mathbb{R}^d$ is a collection of d linearly independent vectors that span $\mathbb{R}^d$. This is the most efficient spanning set. Given that the dimension of $\mathbb{R}^d$ is d, the basis must contain exactly d vectors.

2.2 Inner Product

The Euclidean inner product between two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$ is defined as:

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^{d} a_i b_i$$

This can be represented using different notations such as $\langle \mathbf{a}, \mathbf{b} \rangle$, $\mathbf{a} \cdot \mathbf{b}$, or $\mathbf{a}^\top \mathbf{b}$. The inner product of two vectors is zero if and only if the vectors are perpendicular.

2.3 Outer Product

The outer product of two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$ is given by:

$$\mathbf{a}\mathbf{b}^\top$$

This results in a $d \times d$ matrix, which is different from the scalar result of the inner product.

3 Norms and Properties

3.1 ℓ_p Norm

The ℓ_p norm of a vector $\mathbf{v} \in \mathbb{R}^d$ is defined as:

$$\|\mathbf{v}\|_p = \left(\sum_{i=1}^{d} |v_i|^p \right)^{\frac{1}{p}}$$

where $p \geq 1$. For $p = \infty$, the ℓ_∞ norm is defined as the maximum absolute value of the vector's entries:

$$\|\mathbf{v}\|_\infty = \max_i |v_i|$$

The ℓ_1 norm and the ℓ_∞ norm are dual norms. The ℓ_2 norm, also known as the Euclidean norm, is its own dual. The ℓ_0 norm, defined as the number of non-zero entries in the vector, is not a norm because it does not satisfy the triangle inequality.

3.2 Norm Definition

A norm is a function $\|\cdot\| : \mathbb{R}^d \to \mathbb{R}$ that satisfies the following properties:

1. **Positivity:** $\|\mathbf{v}\| \geq 0$ with $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = 0$.
2. **Homogeneity:** For any scalar $\alpha \in \mathbb{R}$, $\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\|$.
3. **Triangle Inequality:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Norms satisfying these properties are convex functions, while pseudo-norms like the ℓ_0 norm are not.

3.3 Cauchy-Schwarz Inequality

The Cauchy-Schwarz inequality relates the inner product of two vectors to their norms:

$$|\langle \mathbf{a}, \mathbf{b} \rangle| \leq \|\mathbf{a}\|_2 \|\mathbf{b}\|_2$$

This inequality generalizes to other norms, such as:

$$|\langle \mathbf{a}, \mathbf{b} \rangle| \leq \|\mathbf{a}\|_p \|\mathbf{b}\|_q \quad \text{where } \frac{1}{p} + \frac{1}{q} = 1$$

For random vectors $\mathbf{a}$ and $\mathbf{b}$, we have:

$$\mathbb{E}\left[\langle \mathbf{a}, \mathbf{b} \rangle^2\right] \leq \mathbb{E}\left[\|\mathbf{a}\|_2^2\right] \mathbb{E}\left[\|\mathbf{b}\|_2^2\right]$$

4 Angles and Projections

Given two vectors $\mathbf{a}$ and $\mathbf{b}$, the angle θ between them can be determined through the cosine of the angle:

$$\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} \tag{1}$$

The projection of a vector $\mathbf{a}$ onto another vector $\mathbf{b}$ is denoted by $\text{Proj}_{\mathbf{b}}(\mathbf{a})$ and is defined as:

$$\text{Proj}_{\mathbf{b}}(\mathbf{a}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2} \mathbf{b} \tag{2}$$

In the equation above, the term $\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2}$ represents the magnitude of the projection, while $\mathbf{b}/\|\mathbf{b}\|$ represents the direction of the projection.

Alternatively, the projection can also be expressed as:

$$\text{Proj}_{\mathbf{b}}(\mathbf{a}) = \|\mathbf{a}\| \cos(\theta) \frac{\mathbf{b}}{\|\mathbf{b}\|} \tag{3}$$

In the context of geometric interpretation, this figure can help visualize the angle and projection between vectors:

5 Proof of the Cauchy-Schwarz Inequality

The Cauchy-Schwarz inequality, which relates the inner product of two vectors to their norms, states that:

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\|\|\mathbf{b}\| \tag{4}$$

The proof of the Cauchy-Schwarz inequality can be constructed using the concept of angles and projections, as discussed above. Specifically, by considering the projection of vector $\mathbf{a}$ onto vector $\mathbf{b}$ and utilizing the Pythagorean Theorem, one can arrive at the inequality.

6 Projections and Related Concepts

The concept of projection can be illustrated as follows: Consider a point $\mathbf{u}$ that is completely orthogonal to a subspace S. In this case, the projection of $\mathbf{u}$ onto S will be zero. On the other hand, if you have another point $\tilde{\mathbf{u}}$, the projection of $\tilde{\mathbf{u}}$ onto S can be found by drawing a perpendicular (right) angle from $\tilde{\mathbf{u}}$ to S. The length of this perpendicular line represents the projection of $\tilde{\mathbf{u}}$ onto S.

This brings us to the concept of the orthogonal complement. If I give you a subspace S, you can find another subspace that contains all vectors orthogonal to S. Orthogonal here means that the projection of those orthogonal vectors onto S is zero.

Another related concept is the affine subspace. This concept involves shifting a subspace along the direction of an arbitrary vector $\mathbf{a}$. Given a non-zero vector $\mathbf{a}$ and a linear subspace S, the set $\mathbf{a} + S$ is defined as an affine subspace.

7 Projection onto a Convex Set

A set is convex if, for any two points in the set, the line segment connecting them lies entirely within the set. Conversely, a set is non-convex if this condition does not hold.

Now, suppose you have a convex set and want to find the projection of a particular vector onto this convex set. Finding this projection generally involves solving a convex optimization problem. Although this problem may not have a closed-form solution, it can be solved efficiently using optimization software.

The optimization problem for finding the projection of a vector $\mathbf{u}$ onto a convex set C is as follows:

$$\min_{\mathbf{w} \in C} \|\mathbf{w} - \mathbf{u}\|_2 \tag{5}$$

Here, you seek the vector $\mathbf{w}$ within the convex set C that is closest to $\mathbf{u}$ in terms of Euclidean distance.

One useful property of projection is the concept of contraction. Specifically, if you have two points and find their projections onto a convex set, the distance between the projections is not greater than the original distance between the points:

$$\|\text{Proj}_C(\mathbf{u}_1) - \text{Proj}_C(\mathbf{u}_2)\| \leq \|\mathbf{u}_1 - \mathbf{u}_2\| \tag{6}$$

8 Useful Inequalities

We will now review a few useful inequalities.

8.1 Triangle Inequality

The triangle inequality states that for any vectors $\mathbf{a}$ and $\mathbf{b}$, the norm of their sum is less than or equal to the sum of their norms:

$$\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\| \tag{7}$$

This inequality holds for any norm, not just the Euclidean norm.

8.2 Law of Cosines

The law of cosines, which is essentially a quadratic expansion, states that:

$$\|\mathbf{a} + \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 + 2(\mathbf{a} \cdot \mathbf{b}) \tag{8}$$

This is called the law of cosines because, if we normalize by the norms of $\mathbf{a}$ and $\mathbf{b}$, the inner product term becomes the cosine of the angle θ between $\mathbf{a}$ and $\mathbf{b}$.

8.3 Young's Inequality

Young's inequality provides a bound for the inner product of two vectors:

$$\mathbf{a} \cdot \mathbf{b} \leq \frac{\|\mathbf{a}\|^2}{2\lambda} + \frac{\lambda\|\mathbf{b}\|^2}{2} \tag{9}$$

This inequality holds for any positive scalar λ. The proof of Young's inequality relies on the quadratic expansion and the Cauchy-Schwarz inequality.

Useful Identity:

$$\langle a, b\rangle = \frac{\|a\|^2 + \|b\|^2 - \|a - b\|^2}{2}$$

8.4 Generalization of Triangle Inequality

A generalization of the triangle inequality, connected to convexity, states that for any vectors $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$:

$$\left\|\sum_{i=1}^{n} \mathbf{a}_i\right\|^2 \leq n \sum_{i=1}^{n} \|\mathbf{a}_i\|^2 \tag{10}$$

This inequality is useful in the context of optimization, particularly in deep learning, where the loss function often involves a summation structure (e.g., empirical risk minimization).

9 Matrices

Let us briefly discuss some basic concepts related to matrices. A matrix M is an $n \times d$ array of numbers, where n represents the number of rows and d represents the number of columns. The (i, j) entry of matrix M is denoted by M_{ij}.

9.1 Null Space and Range

The null space of a matrix M is the set of all vectors $\mathbf{x}$ such that $M\mathbf{x} = \mathbf{0}$. The range of a matrix M is the set of all vectors $\mathbf{y}$ that can be written as $M\mathbf{x}$ for some vector $\mathbf{x}$.

9.2 Rank

The rank of a matrix is the dimension of the subspace spanned by its columns (or rows). For an $n \times d$ matrix M, the rank is less than or equal to the minimum of n and d.

For a square matrix (i.e., $n = d$), the rank of the matrix is d if and only if the following conditions are equivalent:

- The determinant of M is non-zero.
- The null space of M contains only the zero vector.
- The range of M is the entire space $\mathbb{R}^d$.
- The columns (or rows) of M are linearly independent.

9.3 Eigenvalues and Eigenvectors

The concept of eigenvalues and eigenvectors is crucial when dealing with square matrices. For a square matrix M, an eigenpair $(\mathbf{v}, \lambda)$ satisfies:

$$M\mathbf{v} = \lambda\mathbf{v} \tag{11}$$

Here, λ is the eigenvalue and $\mathbf{v}$ is the corresponding eigenvector.

To find the eigenvalues, you solve the characteristic equation:

$$\det(M - \lambda I) = 0 \tag{12}$$

Once the eigenvalues are found, the eigenvectors can be determined by solving the corresponding linear system of equations.

For symmetric matrices, all eigenvalues are real, while for non-symmetric matrices, the eigenvalues may be complex. A simple example is the matrix:

$$M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

The eigenvalues of this matrix are purely imaginary, $\pm i$. You are encouraged to find these eigenvalues by solving the characteristic equation.

10 Eigenvalues and Spectral Decomposition

For this course, when we deal with eigenvalues, we will primarily focus on symmetric matrices. A common example of such a matrix is the Hessian of a loss function. From the concept of eigenvalues, we can define what is known as the **eigenvalue decomposition** or **spectral decomposition** of a matrix.

Theorem 1. *Let M be a symmetric matrix, i.e., $M \in \mathbb{R}^{d\times d}$ and $M = M^\top$. Then M can be decomposed as:*

$$M = \sum_{i=1}^{d} \lambda_i v_i v_i^\top$$

where λ_i are the eigenvalues and v_i are the corresponding eigenvectors of M. This can also be written in matrix form as:

$$M = V\Lambda V^\top$$

where V is a matrix whose columns are the eigenvectors v_i, and Λ is a diagonal matrix containing the eigenvalues λ_i.

You might also have heard of the **singular value decomposition (SVD)**. It is important not to confuse spectral decomposition with singular value decomposition.

Singular value decomposition applies to a general matrix $A \in \mathbb{R}^{m\times n}$ (which need not be symmetric or square) and states that A can be decomposed as:

$$A = U\Sigma V^\top$$

where U is an $m \times m$ matrix whose columns are the left singular vectors, Σ is an $m \times n$ diagonal matrix with singular values on the diagonal, and V is an $n \times n$ matrix whose columns are the right singular vectors.

The relationship between singular values and eigenvalues is given by:

$$\sigma_i(A) = \sqrt{\lambda_i(A^\top A)}$$

where $\sigma_i(A)$ are the singular values of A, and $\lambda_i(A^\top A)$ are the eigenvalues of $A^\top A$.

Note that if A is symmetric (and hence square), the singular values are the absolute values of the eigenvalues, and the matrices U and V are identical.

Summary:

- Singular values are always non-negative, while eigenvalues can be negative.
- Eigenvalues are defined only for square matrices, while singular values are defined for any matrix.

11 Positive Semi-Definite (PSD) Matrices

A matrix $M \in \mathbb{R}^{d\times d}$ is called **positive semi-definite (PSD)** if:

- M is symmetric, i.e., $M = M^\top$, and
- All its eigenvalues are non-negative.

This is equivalent to saying that the quadratic form $x^\top Mx \geq 0$ for any vector $x \in \mathbb{R}^d$.

The notation $M \succeq 0$ indicates that M is PSD.

To understand why these conditions are equivalent, we can use the spectral decomposition of M:

$$x^\top Mx = \sum_{i=1}^{d} \lambda_i (v_i^\top x)^2$$

Since $\lambda_i \geq 0$ and $(v_i^\top x)^2 \geq 0$, the entire sum is non-negative.

Why do we care about PSD matrices? Because the Hessian of a function at a local minimum is a symmetric matrix, and for a convex function, the Hessian is PSD everywhere.

Examples of PSD Matrices: If $A \in \mathbb{R}^{m \times n}$ (not necessarily symmetric or square), then both $AA^\top$ and $A^\top A$ are PSD matrices. This follows directly from the quadratic form:

$$x^\top AA^\top x = \|A^\top x\|_2^2 \geq 0$$

which is non-negative.

12 Trace and Determinant

Given a square matrix $M \in \mathbb{R}^{d \times d}$:

- The **trace** of M, denoted $\mathrm{tr}(M)$, is the sum of the diagonal elements of M. Interestingly, the trace is also equal to the sum of the eigenvalues of M.
- The **determinant** of M is equal to the product of its eigenvalues.

Matrix Inner Product:

The inner product of two matrices $A, B \in \mathbb{R}^{m \times n}$ can be defined as:

$$\langle A, B \rangle = \mathrm{tr}(A^\top B)$$

This generalizes the Euclidean inner product to matrices and results in a scalar.